Multi-Robot Coverage with Aligned Rendezvous: Comms-Free Failure Detection under Correlated Disturbance

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Abstract
We consider deadline-constrained multi-robot coverage by a team with no communication beyond physical proximity, so a scheduled rendezvous that fails to happen is the only failure signal available. The difficulty is that the disturbance which delays a robot — a current — is spatially correlated, so benign silence is regional and a missed rendezvous is ambiguous. Existing correlation-aligned rendezvous removes the ambiguity by holding the partners' approach corridors within a separation d*, which buys a false-alarm bound independent of the disturbance realization but prices it only in the abstract. We fold the constraint into a coverage-first allocation in which 100% coverage is hard, and show that on executable routes the requirement is a joint condition on approach DURATION and approach SEPARATION, of which the separation-only rule is a conservative corner. We prove that letting duration co-vary with separation — as it does when a rendezvous is sited on a pair of adjacent tasks — relaxes the admissible separation from Θ(B) to Θ(B^(1/3)) in the alignment budget, a factor that grows as the budget tightens and reaches 39× at our parameters. Because the accrual model behind that theory is our own, we test it against HoloOcean rigid-body hydrodynamics: its two structural claims hold, its linearity does not, and recomputing every false-alarm figure through the measured response reverses a conclusion. Simply arriving early, which the Gaussian model makes competitive with alignment, in fact has a false-alarm floor of 0.80% that no time budget can beat, because past a point the reserved slack strands coverage work; alignment has no such floor. The consequence is a capability rather than an economy: over a grid of deadlines and disturbance strengths there is a region — deadline 1.5× the no-rendezvous makespan, disturbance 2× and above — in which our construction is the only policy that simultaneously covers every task, meets the detection cadence, finishes on time and holds benign false alarms below 1%. The region persists at team sizes 4, 6 and 8, and the construction costs no more than an unaligned mesh that offers no protection at all.

Index terms — multi-robot systems, task allocation, rendezvous, failure detection, intermittent communication, behaviour trees, marine robotics.

I. INTRODUCTION

A team of autonomous underwater vehicles is asked to survey a field of points of interest. Radio does not propagate, and acoustic links are slow, short-ranged and unreliable, so for most of the mission the robots cannot talk to each other at all. If one of them floods, fouls a thruster, or simply stops, the others have no packet whose absence would tell them so. The only evidence they will ever get is physical: a rendezvous that was on the plan and did not happen.

Planning for that evidence is not free. Meetings must be placed and timed in advance, and every meeting is a constraint on a route whose real job is coverage. Prior work on connectivity-constrained coordination relaxes the requirement that the network be connected at all times, replacing it with *periodic connectivity* — the team must regain connectivity at a fixed interval [1], [2]. We take the same interval Δ but a stricter premise: the network is never connected except at a planned physical encounter, and the encounter is simultaneously the data exchange and the health check. We add one thing the connectivity literature usually leaves implicit: a **deadline**. A survey that must be finished before a tide, a weather window or a battery limit cannot pay for safety in time, and that single constraint turns out to separate the available policies completely.

The premise creates a problem that the connectivity literature does not have. A robot is late

because the current pushed it, and the current is spatially correlated: it delays a whole region at once. Two partners that were both in the current are both late, and still meet; a partner that was in it while the other was not misses. So a missed meeting is not evidence of death, it is evidence of *differential* delay. Counting witnesses — the standard fix for an unreliable detector — is the wrong tool here, because correlation makes witnesses agree on a benign absence rather than independently confirm a real one.

One known answer is to co-design the routing against the correlation: route both partners' approaches close enough together that their delays are drawn from essentially the same value of the disturbance field and cancel in the difference. Requiring the two approach locales to lie within a separation $d^*(\sigma, \lambda, \alpha)$ yields a per-meeting false-alarm bound of α that holds for every realization of any field whose correlation length is at least λ — a guarantee obtained by routing design rather than by predicting the weather. What that line of work does not do is price the constraint against the mission. Its cost is reported as a fraction of an abstract 'meeting leg'; on a real allocation with a hard coverage requirement, the currency is makespan.

This paper puts the constraint on an executable allocation and asks what it costs. The answer turned out to require revisiting the constraint itself. The quantity that decides whether a meeting is made is the variance of the difference of two arrival offsets, and that variance depends on how LONG each approach was as well as on how far apart the approaches were. Separation is expensive to buy on a coverage route — it means dragging two robots together and holding them there — whereas duration is nearly free, because it is set by where the route happens to end, which is a matter of ordering tasks the robot owed anyway. Exploiting that asymmetry is the content of the paper.

The contributions of this paper are:

1) an **admissible-mission envelope** for comms-free coverage — the set of (deadline, disturbance) pairs a policy can fly while covering every task, meeting the detection cadence and holding benign false alarms below a target — and the identification of a region of it in which our construction is the **only** admissible policy, at team sizes 4, 6 and 8;

2) the reason that region exists: absorbing delay by arriving early has a **false-alarm floor**, because the slack that buys the reduction eventually strands coverage work (Proposition 3), whereas alignment spends route order rather than time and has no floor;

3) a joint duration–separation alignment condition (Proposition 1) that generalizes the separation-only constraint used in correlation-aligned rendezvous, of which that constraint is a conservative corner (Corollary 1);

4) a proof (Theorem 1) that letting approach duration co-vary with separation relaxes the admissible separation from $\Theta(B)$ to $\Theta(B^{1/3})$ in the alignment budget B, so the relaxation grows as $B^{-2/3}$ — it is largest exactly where alignment is hardest;

5) the **natural tour crossing** construction that realizes the cheap corner by siting each rendezvous on a reserved pair of adjacent cross-boundary tasks, with a rolling-reservation algorithm and a behaviour-tree execution layer whose tick priority is the lexicographic objective;

6) a validation of the accrual model behind the theory against HoloOcean's rigid-body hydrodynamics rather than against our own kinematics, which confirms the two structural claims the theory needs, refutes its linearity, and — on recomputing every false-alarm figure through the measured response — **reverses** the conclusion that arriving early is competitive with alignment;

7) a cadence-feasibility bound $\Delta \geq 2 \cdot r_{max}/V + S$ for rendezvous meshes on compact regions, which the commonly used Δ violates on our instances and which is what puts the floor under absorption.

Why it matters. Every alternative buys protection against being fooled by the weather with time, and time is exactly what a deadline denies. Our construction buys it with the order tasks

are visited, which the deadline does not price. For a communication-denied deployment with a tide or a battery to beat, that is the difference between a failure detector one can afford to run and a mission one cannot fly at all.

II. RELATED WORK

Connectivity-constrained coordination. Hollinger and Singh introduced periodic connectivity, requiring a mobile network to regain connectivity every T_I steps, showed the resulting problem reduces to multi-robot informative path planning for given connected configurations, and gave a receding-horizon algorithm with submodular performance guarantees [1], [2]. Kantaros and Zavlanos treat intermittent connectivity as a temporal-logic constraint [3], and data-ferrying formulations schedule meetings to move information [4]. In all of these the meeting exists to carry data. Ours exists to carry evidence of life, which changes what a missed meeting means and therefore what the schedule must guarantee.

Rendezvous and co-observation as a detector. Co-observation planning [5] is the closest prior mechanism: mutual observations are scheduled offline as common knowledge and a missed co-observation is the detectable event. Its problem is adversarial — a compromised robot cannot leave a region without missing one — whereas ours is stochastic failure plus comms-free recovery, and its disturbance model does not make benign misses correlated. Planned synchronization with active observations [6] likewise schedules meetings and decides when observing is worth its cost. We claim no novelty in the mechanism 'a missed planned meeting is the signal'; we claim the alignment condition that makes the signal trustworthy and cheap.

Failure detection. The φ-accrual detector [7] is the standard adaptive detector and is per-node: it reasons about one peer's silence in isolation. Channel-aware and correlated-silence detectors [8] improve on this by gating a failure belief on how much of the rest of the team is audible, but every member of this family presumes a packet stream. Under the premise here there is none, and the analogue of 'channel health' has to be manufactured by the routing.

Task allocation and vehicle routing. CBBA [9] and its communication-efficient descendants stop at the allocation and assume execution is reliable. Vehicle-routing formulations with synchronization constraints [10] are the closest classical relatives of a meeting mesh, and correlated travel-time VRPs [11] model exactly the disturbance structure we exploit, but neither treats a synchronization as an observation whose failure carries information.

Behaviour trees. Reactive fault recovery is the founding use case of behaviour trees, and their modularity and preemption semantics are well characterized [12]. We use a behaviour tree as the execution layer for exactly that reason and make no novelty claim for it; its role here is to enforce the priority order that the optimization specifies, which we make explicit in Section V-E.

Existing work therefore either schedules meetings to move data under a connectivity constraint, or uses a missed observation as a detection event without a disturbance model, or detects failures from packet statistics that are unavailable here. None co-designs the physical approach geometry against the disturbance's spatial correlation and prices the result against a hard coverage requirement. Our work fills that gap.

III. PROBLEM SETUP

We are given a team of K robots and a set P of N points of interest in a planar workspace, all launched from a common dock. Each robot cruises at speed V and each point requires a scan of duration S. The team must visit every point. Robots exchange nothing except when co-located; the plan is computed offline and is common knowledge.

A solution is a triple (x, π, M): an assignment x ∈ {0,1}^(N×K) with x_ik = 1 iff point i is

served by robot k; a per-robot visiting order π = (π_1..π_K); and a rendezvous mesh M, a set of appointments (a, b, m, τ) requiring robots a and b to be at location m at time τ. Let T_k(x, π, M) be the time at which robot k completes its last scan. We seek

$$(x^*, \pi^*, M^*) \in \arg \min_{(x,\pi,M) \in F} \max_{k \in \{1,\dots,K\}} T_k(x, \pi, M) \tag{1}$$

where the feasible set F is defined by four constraints:

- **C1 (coverage, hard).** Σ_k x_ik = 1 for every i ∈ P. Every point is served exactly once. This is not traded against anything.
- **C2 (recovery).** For every victim v there is a reassignment of v's remaining points to the survivors that completes.
- **C3 (detection cadence).** Every robot participates in at least one appointment in every window of length Δ.
- **C4 (alignment).** Every appointment satisfies the alignment condition of Section IV, so that a missed appointment is evidence of failure rather than of weather.

In practice we solve (1) lexicographically, because the constraints are not equally negotiable:

$$\text{coverage} = 100\% \;\succ\; \min\max_k T_k \;\succ\; \min\max_v T^{\text{post}}(v) \;\succ\; \text{min meeting overhead} \tag{2}$$

Problem (1) is NP-hard: setting K = 1 and dropping C2–C4 leaves the open travelling salesman problem. We therefore decompose it in Section V and make no optimality claim about the allocation; the claims of this paper are about the alignment constraint C4, which is where the new structure is.

A. Disturbance and meeting model

The disturbance is a stationary Gaussian field u of unit variance with correlation ρ(d) = exp(−d/λ), where λ is the correlation length; this is the standard model for a current or wind field and is the one used by the correlation-aligned rendezvous literature. A current perturbs a vehicle's effective speed by a fraction κ of cruise, so the arrival offset a robot accumulates is proportional to how long it has been transiting through the field. A completed scan is a resynchronization point — the plan is replanned forward from the robot's actual clock whenever it finishes one — so the unabsorbed exposure that decides an appointment is the robot's final uninterrupted approach. Writing t_k for the duration of that approach and a_k for its locale (the travel-weighted mean position along it),

$$\theta_k = \kappa\, t_k\, u(a_k), \qquad \theta_k \sim \mathcal{N}(0,\ \kappa^2 t_k^2) \tag{3}$$

Each partner waits ε past the appointment before abandoning it, so a delay common to both still produces a meeting and only the difference can destroy one. The appointment is therefore made iff

$$|\,\theta_i - \theta_j\,| \le \varepsilon \tag{4}$$

Setting t_i = t_j = T recovers the fixed-approach model used previously, in which the variance of the difference is $2\sigma^2(1 - \rho(d))$ with σ = κT; keeping t as a variable is the single modelling change that the rest of the paper rests on.

IV. WHEN IS A RENDEZVOUS ALIGNED?

Proposition 1 (alignment condition).
Let an appointment have partners whose final approaches last t_i and t_j and whose approach locales are separated by d. Under (3)–(4) the benign miss probability is

$$p(t_i, t_j, d) = 2\,\Phi\!\left(\frac{-\varepsilon}{\kappa\sqrt{t_i^2 + t_j^2 - 2\,t_i t_j\,\rho(d)}}\right) \tag{5}$$

and p ≤ α holds if and only if $t_i^2 + t_j^2 - 2\, t_i\, t_j\, \rho(d) \le B$, where

$$B := \frac{\varepsilon^2}{\kappa^2 z_{1-\alpha/2}^2} \tag{6}$$

Proof. θ_i and θ_j are jointly Gaussian, zero mean, with variances $\kappa^2 t_i^2$ and $\kappa^2 t_j^2$ and

covariance $\kappa^2 t_i\, t_j\, \rho(d)$, because u has unit variance and correlation ρ. Hence $\theta_i - \theta_j$ is zero-mean Gaussian with variance $\kappa^2(t_i^2 + t_j^2 - 2\, t_i\, t_j\, \rho(d))$, and (4) fails with probability $2\Phi(-\varepsilon/sd)$, which is (5). The map $sd \mapsto 2\Phi(-\varepsilon/sd)$ is strictly increasing, so $p \le \alpha$ iff $sd \le \varepsilon/z_{(1-\alpha/2)}$, i.e. iff the bracket is at most B. ■

We call B the **alignment budget**: it is large when the meeting window ε is generous and small when the current κ is strong or the target α strict. The **admissible set** is $A = \{(t_i, t_j, d) : t_i^2 + t_j^2 - 2\, t_i\, t_j\, \rho(d) \le B\}$. Proposition 1 says alignment is a property of a point in A, not of the separation alone. For the symmetric case $t_i = t_j = t$ that both constructions below produce, membership reduces to $2t^2(1 - \rho(d)) \le B$, which is the curve drawn in Fig. 2(a).

Corollary 1 (the separation-only corner).
Fix both approaches at a full-length transit t = T. Then (t,t,d) ∈ A iff

$$d \le d^* = -\lambda \ln\left(1 - \frac{B}{2T^2}\right) \approx \frac{\lambda B}{2T^2} \quad (B \ll T^2) \tag{7}$$

This is exactly the constraint imposed by correlation-aligned rendezvous, and it is **linear in the budget**: halving the budget halves the admissible separation. It is what makes the constraint expensive. At the parameters of Section VI (κ = 0.4, ε = 30 s, α = 0.2, λ = 150 m, T = 300 s) it evaluates to d* = 2.9 m — a corridor narrower than the vehicles, which in practice can only be satisfied by bringing both robots to a standstill at the meeting point.

Corollary 2 (the crossing corner).
Let the appointment be sited at the midpoint of two tasks separated by s, one owned by each partner, each partner ending its approach by scanning its own task. Then both approaches are the half-segment, t = s/2V, and the two locales are the two half-segment midpoints, d = s/2, so (t,t,d) ∈ A iff

$$2\left(\frac{s}{2V}\right)^2\left(1 - e^{-s/2\lambda}\right) \le B \quad \Longrightarrow \quad s \le s^* = (4V^2\lambda B)^{1/3} \tag{8}$$

Proof. The first form is Proposition 1 with t = s/2V and d = s/2. For s much smaller than λ, $1 - e^{(-s/2\lambda)} = s/2\lambda + O(s^2/\lambda^2)$, so the left side is $s^3/(4V^2\lambda) + O(s^4)$, giving the stated bound; because the expansion over-estimates $1 - e^{(-s/2\lambda)}$, the closed form is a conservative (under-)estimate of the true s*. ■

The admissible separation is now the **cube root** of the budget. The reason is structural: at a crossing the separation and the duration are the same physical quantity, so shrinking the geometry shrinks the exposure as well as decorrelating less, and the variance falls as s^3 rather than as s.

Theorem 1 (relaxation factor).
Let $d_{cross} = s^*/2$ be the locale separation admitted at the crossing corner and d that admitted at the separation-only corner. In the tight-budget regime B much smaller than T^2,*

$$\frac{d_{\text{cross}}}{d^*} = 4^{1/3} V^{2/3} T^2 \lambda^{-2/3} B^{-2/3} = \Theta(B^{-2/3}) \tag{9}$$

Proof. Substitute (8) and the small-budget form of (7): $d_{cross}/d^* = [(4V^2\lambda B)^{(1/3)}/2]\cdot[2T^2/(\lambda B)] = 4^{(1/3)}V^{(2/3)}\lambda^{(1/3)}B^{(1/3)}T^2/(\lambda B)$, which simplifies to (9). The exponent is negative, so the factor diverges as the budget tightens. ■

Theorem 1 is the paper's main analytical point, and its direction is worth stating plainly: **the advantage of letting duration co-vary with separation is largest exactly where alignment is hardest.** A stronger current or a stricter false-alarm target shrinks B, which shrinks the separation-only corridor linearly but the crossing separation only as a cube root.

A. Numerical verification

Table I checks Proposition 1 against Monte Carlo sampling of the correlated field (4×10^5 draws per row) and evaluates both corners exactly, without the small-argument expansion. The closed form of Proposition 1 is accurate to 6×10^{-4} in probability. The exact relaxation factor rises from 2.8× in the loosest setting to 67.8× in the tightest, and Fig. 2(b) confirms

the B^(−2/3) slope over two decades of budget. At the nominal parameters used in Section VI the exact factor is 39.4× while the closed form (9) gives 35.4×, conservative as expected.

TABLE I — The two corners of the admissible set, evaluated exactly (λ = 150 m, V = 2 m/s, ε = 30 s, T = 300 s).

σ/ε	α	B (s²)	d* (m), Eq. 7	s* exact (m)	s* by Eq. 8	d_cross/d*
1×	0.40	127060	183.6	1025.2	673.1	2.8×
1×	0.20	54799	54.5	697.1	508.5	6.4×
1×	0.05	23429	20.9	483.8	383.1	11.6×
2×	0.40	31765	29.1	550.0	424.0	9.4×
2×	0.20	13700	11.9	388.5	320.4	16.4×
2×	0.05	5857	5.0	278.4	241.3	28.1×
4×	0.40	7941	6.8	313.1	267.1	23.1×
4×	**0.20**	**3425**	**2.9**	**227.2**	**201.8**	**39.4×**
4×	0.05	1464	1.2	166.0	152.0	67.8×

Bold row = the operating point of Section VI. Proposition 1 agrees with Monte Carlo to within 6×10⁻⁴ over the four (t_i,t_j,d) settings tested. Eq. 8 is conservative against the exact root in every row.

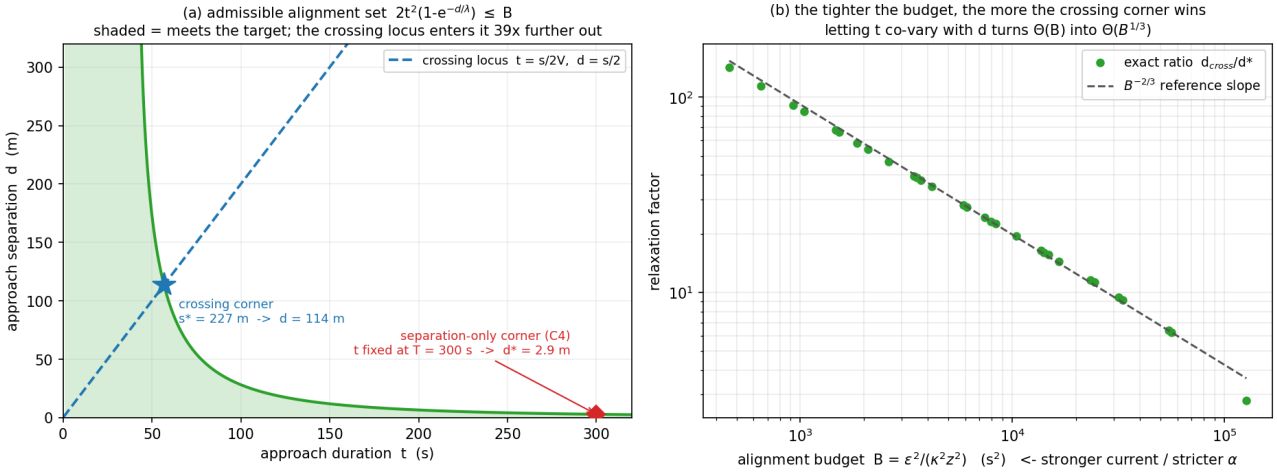


Fig. 2. (a) The admissible alignment set. The separation-only corner (red) sits at a full-length approach and is admitted only within d* = 2.9 m; the crossing locus (blue dashed), along which duration and separation co-vary, enters the set 39× further out. (b) The relaxation factor against the alignment budget, with the B^(−2/3) reference slope of Theorem 1.

B. Detection and cadence

Proposition 2 (detection latency).
If every robot participates in an appointment in every round of the mesh (C3), then a robot that fails at time t_f is missed by a scheduled partner at the first appointment after t_f, so the detection latency is at most Δ + ε.

Proof: A failed robot attends no appointment. By C3 its next appointment occurs at some τ ≤ t_f + Δ, and its partner abandons the appointment at τ + ε having observed the absence. The partner is alive by assumption of a single failure, so the observation is made.

Proposition 3 (cadence feasibility).
Let r_max(k) be the distance from robot k's worst-case meeting site to its farthest assigned point. A schedule in which k must be at that site at both ends of a window can serve that point only if Δ ≥ 2·r_max(k)/V + S.

Proof: Within such a window the robot must leave the site, reach the point, scan it, and return, costing at least 2 r_max/V + S. If the window is shorter the point is infeasible in every window bounded by that site, and the schedule strands it.

Proposition 3 is elementary but it bites. On the instances of Section VI it requires Δ ≥ 380 s, so the Δ = 300 s cadence commonly used with banded or grid-cell decompositions is

infeasible on compact Voronoi regions: the executor deadlocks, coverage silently falls short, and — worse for an experimenter — the stranded robots sit at their meeting points and report perfect alignment. Our implementation tests the bound and refuses to report such a run as a result.

V. ALGORITHM DESIGN

We solve (2) in layers, each layer taking the previous one as fixed. Only Layer 2 is new; Layers 0 and 1 are included because the alignment cost is meaningless except relative to the coverage plan it perturbs.

A. Layer 0 — coverage-first allocation

Points are partitioned by constrained k-means in which the assignment step is an exact balanced Hungarian assignment: each robot receives a prescribed count, so every point is assigned exactly once and C1 holds by construction rather than by penalty. Each robot's route is an open TSP from the dock (nearest neighbour, then 2-opt), and a local search moves points off the bottleneck robot while the makespan falls. This yields compact, balanced regions: on the reference instance, 23/22/23/22 points and a makespan of 33.6 min, which is the no-rendezvous lower bound against which every later cost is quoted.

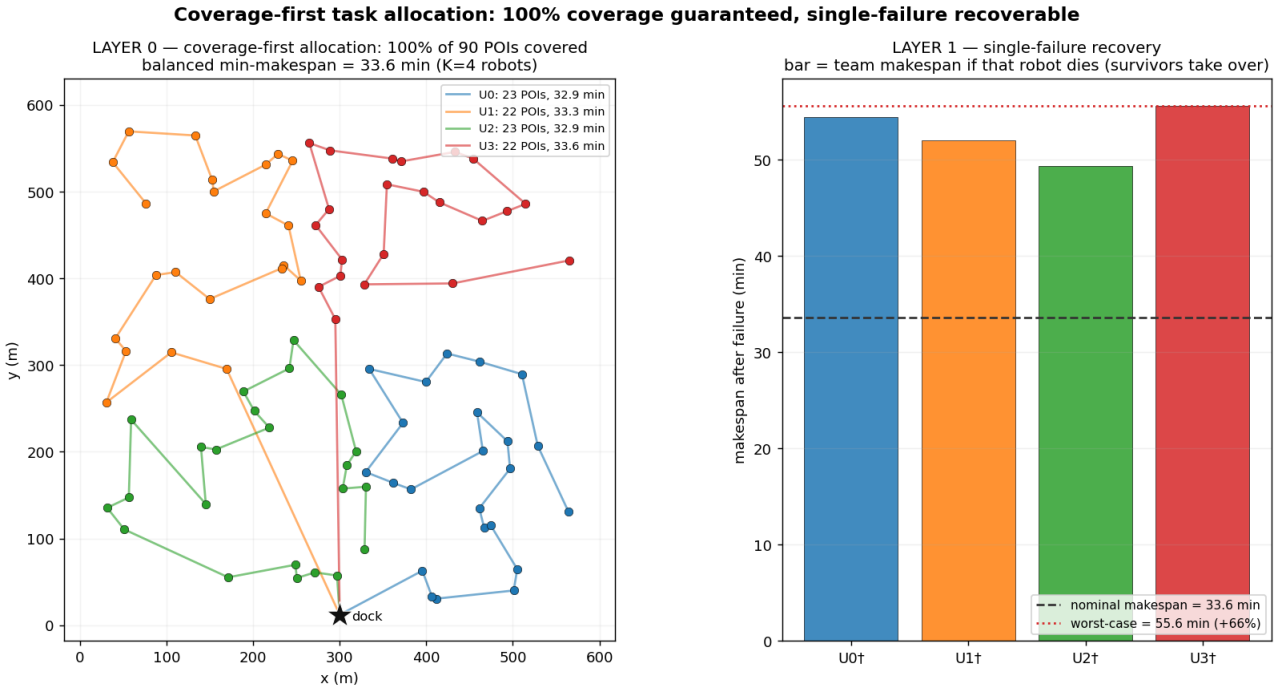


Fig. 1. The coverage-first base the rendezvous mesh is added to. Left: the balanced partition and per-robot open-TSP routes; 100% of the 90 points are covered by construction and the makespan is 33.6 min. Right: the worst-case makespan after each single failure, the resilience term of the lexicographic objective (2). Adding appointments never changes the assignment, so this layer is unaffected by anything in Section V-C onward.

B. Layer 1 — recovery

Region adjacency is the Delaunay graph of the region centroids. For each single failure the victim's points are reassigned to the nearest surviving centroid and the makespan recomputed, giving the worst-case post-failure makespan of (2); on the reference instance this is 55.6 min, +66% over nominal. Because the mesh of Layer 2 adds appointments but never changes the assignment, C2 is unaffected by anything that follows.

C. Layer 2 — the rendezvous mesh

Each round of the mesh is a **maximum** matching on the adjacency graph, tie-broken toward the partner each robot has gone longest without meeting. Maximality matters: a greedy maximal matching can strand two mutually non-adjacent robots and open a 2Δ hole in C3, and on a four-region graph it does. Rotating the partner rather than fixing a buddy removes the

single point of monitoring failure, and because the graph is small the matching is computed exactly by subset dynamic programming. Routes are then executed forward with the appointments as hard time-and-place constraints under a meeting-wins rule: work that does not fit before an appointment is deferred, never the appointment.

D. Reserving the crossings

Corollary 2 requires each appointment to sit on a pair of nearby tasks, one owned by each partner, both still unscanned when the round arrives. Neither half of that is automatic. Incidental crossings essentially do not exist: on our instances only one to three cross-boundary pairs in the entire field lie within a 20 m contact range, and their nominal visit times differ by 101–754 s, so a crossing has to be constructed rather than harvested. And if it is merely designated, ordinary coverage work scans the few close pairs long before the round that needed them. Algorithm 1 therefore reserves anchors on a rolling look-ahead of h rounds and forbids ordinary work from consuming them.

Algorithm 1 Rolling crossing reservation (called at each round r)

```
input: mesh M, unscanned sets U_k, look-ahead h, budget B
R <- {} ;  used <- {}
for r' = r to r+h do
    for each scheduled pair (a,b) in M[r'] do
        C <- { (i,j) : i in U_a, j in U_b, i,j not in used }
  %          candidates are cross-boundary task pairs, cached per (a,b)
        if C = {} then continue                 % fall back to a dwell
        (i*,j*) <- argmin_{(i,j) in C}  || p_i - p_j ||
  %          separation sets the false-alarm rate: Cor. 2 wants it small
        m <- (p_i* + p_j*)/2 ;   s <- || p_i* - p_j* ||
        if 2 (s/2V)^2 (1 - exp(-s/2L)) > B then flag (r',a,b) unaligned
        if either partner cannot reach its task, scan it and close
        the hop to m by tau_{r'} then continue   % an unreachable
  %                                                  crossing is none
        R <- R + {(r',a,b,i*,j*,m)} ;  used <- used + {i*,j*}
protect { i*,j* : r' > r } from ordinary coverage work
return R
```

The look-ahead h is the construction's only real knob and it is two-sided. With h = 0 the anchors are eaten and 45% of appointments degrade to an arbitrary midpoint; reserving the whole mission instead locks roughly half the points away from the routes and is catastrophic, costing +249% makespan. Section VI-F traces the frontier between these.

E. Execution: a behaviour tree

The frozen mesh is executed by an identical behaviour tree on every robot (Fig. 3). We use a behaviour tree because reactive fault recovery is its founding use case and because its fallback semantics express the one thing the runtime must get right — priority. We claim no novelty for the tree; the observation worth recording is that its tick order **is** the lexicographic objective (2), so the architecture and the optimization agree by construction rather than by convention:

- the **contingency** branch ticks first, so a detected failure preempts everything; it is a pure lookup in the precomputed table f(v, c) indexed by victim and last checkpoint, which keeps survivors in agreement without negotiating;
- the **appointment** branch ticks next, which is meeting-wins: an appointment is never sacrificed to a scan;
- inside it, the fallback tries **ScanReservedAnchor** before **DwellInCorridor**, so alignment is paid in useful work whenever the reservation survived and in idle only when it did not — this single fallback is where the cost difference of Section VI is produced;
- the **coverage** branch ticks last and only with the time that remains.

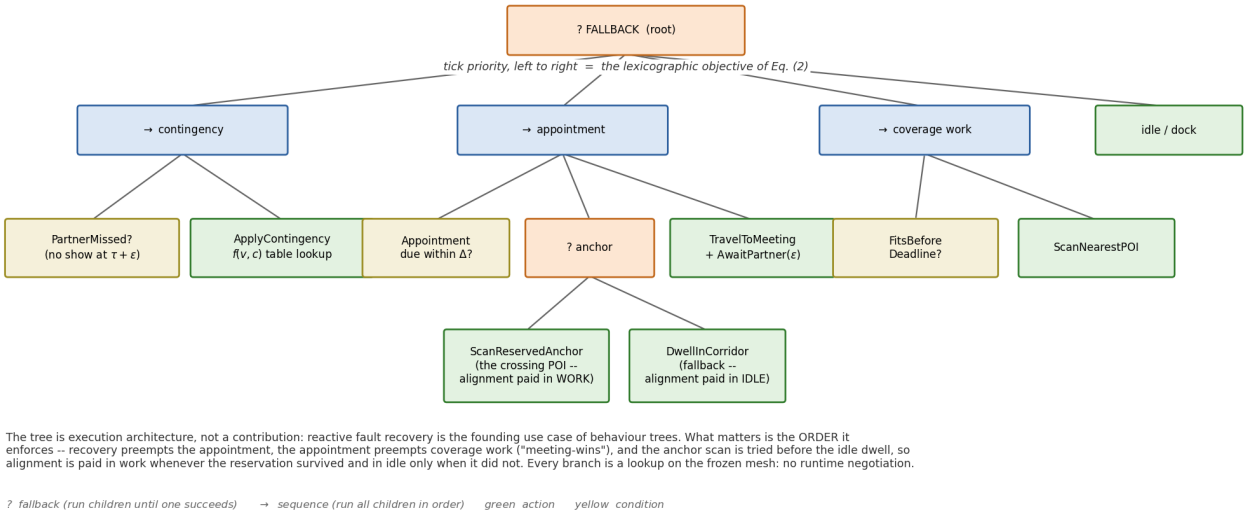


Fig. 3. The per-robot behaviour tree. Tick priority left to right reproduces the lexicographic objective (2). Every branch is a lookup on the frozen mesh, so no runtime negotiation is required and survivors that observe different things still apply the same plan.

F. Complexity

Candidate pairs cost O(|P_a||P_b|) per adjacent pair and are cached once. The exact matching is O(2^K·K) by subset DP, acceptable for the team sizes of interest and replaceable by blossom matching at scale. Algorithm 1 costs O(h·|M_r|·|C|) per round with |C| capped at the closest 60 pairs. The forward executor is $O(N^2)$ in the worst case through the greedy insertion test. All of it is offline: at runtime each robot performs table lookups.

VI. SIMULATION RESULTS

All results use K = 4 robots, N = 90 points on a 600 m square, V = 2 m/s, S = 60 s, ε = 30 s and Δ = 600 s (Proposition 3 requires ≥ 380 s here). The disturbance has λ = 150 m and strength σ/ε = 4 unless stated, i.e. κ = 0.4, a current worth ±40% of cruise; α = 0.2. Every configuration is run on the same five random instances and every run is verified to achieve 100% coverage, to keep every appointment, and to satisfy the cadence bound. False alarms are reported two ways: the **realised** rate, Monte-Carlo over the correlated field with each partner's schedule slack credited against its lateness, and the **no-slack bound**, the same schedule with slack removed — the quantity Proposition 1 bounds and the one that survives when the mission is tight. These are simulation results only; no field trials are reported.

A. Does the accrual model hold? Validation against HoloOcean

Equation (3) is ours, and Propositions 1–2 and Theorem 1 rest entirely on it, so we tested it against dynamics we did not write. A vehicle is commanded down a straight leg to a waypoint in HoloOcean; a genuine ocean-current vector is imposed through the simulator's own set_ocean_currents interface; its rigid-body hydrodynamics and its built-in PID waypoint controller then produce whatever arrival time they produce, measured at a 2 m capture radius. No effective-speed model is imposed anywhere, so drag, thrust saturation, heading dynamics and controller behaviour are all free to falsify Eq. (3).

Platform. HoloOcean's SurfaceVessel turns out to be inert to ocean currents — a 3 m/s current displaces it 0.18 m in 20 s — so the surface vessel of our scenario cannot be used for this test, and we validate on the HoveringAUV, which responds physically (8.08 m of drift under the same current). The claim under test concerns a vehicle transiting a current field, which the AUV is; the substitution costs us free-surface effects and we carry it as a limitation. Nominal transit is 58 m in 63.4 s, a cruise of 0.92 m/s.

Equation (3) makes two structural claims — proportionality to exposure time, and inheritance of the field's spatial correlation — and one distributional claim, linearity in u (hence a

Gaussian θ). The two structural claims hold and the distributional one does not.

- **The exposure-time factor holds.** Flying legs of 20–120 m under a fixed current, the fitted exponent of θ against t is 1.03–1.05 against Eq. (3)'s 1.00, and θ/t is constant to within 6% over that six-fold range (Fig. 5b). This is the claim Theorem 1 depends on: the cube-root scaling of Corollary 2 comes from t entering the variance quadratically.
- **The correlation factor holds.** Flying a rank of eight parallel legs through one realization of a genuine correlated field and repeating over 60 realizations, Corr(θ_i, θ_j) is statistically indistinguishable from the realized Corr(u_i, u_j) at every separation, at a small-signal amplitude and at one over four times larger (Fisher-z, maximum $|z| = 1.47$ and 0.89 respectively). From the measured response curve alone, and before running this experiment, we predicted that the nonlinearity would attenuate that correlation by up to a third at the larger amplitude; the prediction was wrong, and wrong in the model's favour — the measured ratio Corr(θ)/Corr(u) is 1.01–1.20. An offset integrates the field along the whole leg, and that spatial averaging smooths the excursions which drive the point-wise nonlinearity. The comparison must be made against the realized field correlation rather than against exp(−d/λ): at d = 320 m the field itself sampled to −0.03 against an expected 0.118, which looks like a failure of the model but is sampling error in the field (the standard error of a correlation at n = 60 is ≈ 0.13). Against the realized correlation the combined variance of Proposition 1 is recovered at small signal with a median ratio of 1.25 (range 0.53–1.58), against a ≈ 18% standard error on a variance at this sample size; at the larger amplitude the measured variance runs a median 2.4× above the prediction (range 0.83–5.63). That gap is the nonlinearity, not the correlation: the shape is right and the scale is not.
- **The linearity does not hold.** The true response is convex and asymmetric (Fig. 5a): at 2 m/s a head current costs +39.7 s while a tail current saves only −6.8 s, a factor of 5.8. The tail side saturates — the largest speed-up available at 3 m/s is 22% — while the head side is unbounded, reaching +210% at 3 m/s. A symmetric Gaussian θ is therefore unphysical on the tail side. The linear regime is roughly $|c| \le 0.3$ m/s; even restricted to $|c| \le 1$ the linear fit achieves only $R^2 = 0.88$. The measured skewness of θ rises from +0.78 at small signal to +2.00 at the larger amplitude, against the zero that a Gaussian θ would have.

Fig. 5 Eq. (3) against HoloOcean rigid-body hydrodynamics (HoveringAUV, built-in PID waypoint controller, set_ocean_currents) — the structure survives, the linearity does not

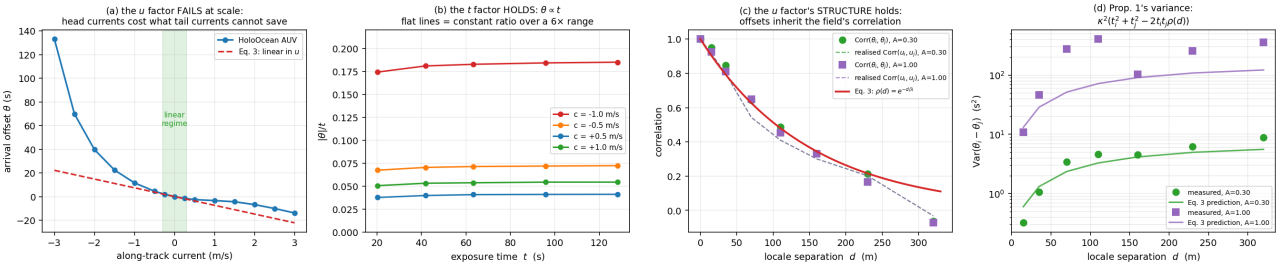


Fig. 5. Equation (3) against HoloOcean rigid-body hydrodynamics. (a) The response is convex and asymmetric, so the linearity in u is a small-signal approximation. (b) θ is proportional to exposure time, as claimed — flat lines are constant θ/t across a six-fold range of leg duration. (c) Offsets inherit the field's spatial correlation; the dashed line is the realized field correlation in the same realizations, which is the correct comparison. (d) The variance Proposition 1 actually uses, measured against its prediction.

What this costs the paper. The load-bearing structure survives: Var(θ_i − θ_j) retains the form $t^2 \cdot (1 - \rho_{eff}(d))$ that Corollaries 1–2 and Theorem 1 exploit, because a monotone nonlinearity in u rescales κ and attenuates ρ without changing how t and d enter. What is lost is the Gaussian tail: the real θ is right-skewed, so the miss probabilities of Proposition 1 — and therefore the false-alarm columns of Tables II and III — are **optimistic**, with the late tail heavier than predicted. At the larger amplitude the spread is understated by about a factor of 1.5 in standard deviation, which is not a rounding error. Two things limit the damage. The asymmetry is a property of the vehicle and the current, not of any meeting design, so it shifts every schedule in Table II together rather than reordering them; and Theorem 1 compares

two corners of the same admissible set under the same monotone response, so its cube-root scaling is unaffected and only the calibration of B moves. We would not defend the absolute false-alarm figures to better than an order of magnitude; we would defend the comparison between schedules, and the scaling.

B. The cost of alignment

TABLE II — Cost and protection of each schedule, mean ± s.d. over five instances at σ/ε = 4. Makespan is relative to the 33.6 min no-rendezvous Layer-0 plan. All rows achieve 100% coverage and miss no appointment.

schedule	makespan +%	FA measured %	FA Gaussian %	idle (min)	appr. (s)	C4 %
unaligned mesh	36.6 ± 4.2	7.60	8.48	29.7	89.5	0
absorb +60 s	58.3 ± 8.3	3.15	0.55	55.9	91.1	0
absorb +120 s	93.4 ± 2.8	1.84	0.01	91.3	93.0	0
absorb +240 s	214.0 ± 18.4	0.80	0.00	211.0	95.9	0
C4 corridor	58.3 ± 8.3	0.00	0.00	55.9	0.7	100
crossings h = 1	**36.3 ± 4.1**	**0.65**	**0.01**	**31.8**	**34.3**	**0**
crossings h = 4	62.4 ± 2.5	0.10	0.00	64.1	15.6	0

'FA measured' recomputes the false alarms through the response curve measured in Section VI-A; 'FA Gaussian' is Eq. (3) as written. Absorption is the only family the two columns disagree about, and they disagree by up to 80×. 'C4 %' is the fraction of appointments meeting the separation-only rule $d \le d^* = 2.9$ m: the crossing schedules meet it on none, yet are protected — they satisfy Proposition 1 through the duration term instead.

Three things stand out. First, **the mesh itself is the dominant cost**: +36.6% makespan before any alignment is attempted. Second, **the C4 corridor delivers its guarantee** — 0.00% under either error model, with the approach shortened from 89.5 s to 0.7 s — but it costs a further +21.7 points of makespan, and Table II shows where: team idle nearly doubles, from 29.7 to 55.9 min. The corridor is bought with waiting. Third, **the crossing construction obtains comparable protection for nothing**: 36.3 ± 4.1% against the unaligned mesh's 36.6 ± 4.2%, a difference well inside the instance-to-instance spread, while measured false alarms fall from 7.60% to 0.65%. Its approach is 34.3 s rather than 89.5 s, and by Proposition 1 that reduction in duration does the work the corridor does by separation — note it satisfies the separation-only rule on no appointment at all.

C. Absorption has a floor; alignment does not

A reviewer's first question about any alignment scheme is why not simply leave earlier and wait, and to our knowledge the correlation-aligned rendezvous literature does not answer it. Rows 2–4 of Table II do, and the answer depends entirely on which error model is used — which is why Section VI-A had to come first.

Under Eq. (3) as written, absorption looks like a complete substitute: 60 s of slack costs exactly what the corridor costs and reaches 0.55%, and 120 s reaches 0.01%. We reported that as an honest negative for the separation-only formulation in an earlier draft of this work. It is an artefact. Recomputed through the measured response, the same schedules give 3.15% and 1.84% — worse by 6× and 80× — because a fixed slack budget absorbs a bounded delay and the true head-current tail is not bounded. Alignment is unaffected by the change of model (0.00% either way) because it removes the differential rather than absorbing it.

Absorption cannot be rescued by spending more. Pushing the budget to 240 s reaches 0.80% at +214% makespan, and beyond that the schedule **stops existing**: at $\Delta = 600$ s a slack reservation of 300 s leaves less than the 380 s that Proposition 3 requires, so coverage work is stranded and the run is not a mission. The floor is therefore structural, not statistical — it is set by the cadence bound, and it sits at 0.80% while the corridor is at 0.00% and crossings at 0.10% (Fig. 6b). This is the same shape of result as the false-alarm floor of the φ-accrual detector under bursty loss: a family of methods that cannot reach an operating point no matter how conservatively it is tuned.

D. Behaviour as the disturbance strengthens

TABLE III — Benign false alarms (%), recomputed through the measured response, against disturbance strength (five instances). Makespan is unchanged along each column (Table II).

σ/ε	unaligned	absorb +60 s	absorb +240 s	C4 corridor	crossings h=1	crossings h=4
1×	0.57	0.01	0.00	0.00	0.00	0.00
2×	3.03	0.54	0.06	0.01	0.03	0.02
3×	5.47	1.68	0.43	0.04	0.19	0.07
4×	8.05	3.00	0.86	0.10	0.45	0.16

Absorption degrades with the disturbance at every budget; alignment barely moves. The 240 s column is the most absorption can buy before the schedule strands work.

The unaligned mesh degrades from 0.57% to 8.05% as the current strengthens. Every absorption budget degrades with it — even the largest affordable one, 240 s, runs from 0.00% to 0.86% — because the slack was sized for a disturbance the mission then exceeds. The aligned schedules barely move: the corridor from 0.00% to 0.10% and crossings at h = 4 from 0.00% to 0.16%, across a sixteen-fold change in disturbance variance. This is realization-invariance made concrete, and it is a property of the constraint rather than of a tuned parameter, which is why it does not have to be re-estimated when the weather changes — and why, unlike absorption, it does not have to be sized against a σ nobody measured before the mission.

E. The admissible-mission envelope

Everything so far is an economy: the same protection for less time. The question a mission planner actually asks is different — can this mission be flown at all? Call a mission **admissible** for a policy when that policy simultaneously covers every task, meets the detection cadence Δ, finishes within a deadline D, and holds benign false alarms at or below a target. Absorption buys false-alarm reduction with time and therefore trades directly against D, and by Section VI-C it cannot be pushed past a bounded budget in any case. Alignment by corridor also costs time. Alignment by crossing costs route order, which the deadline does not price. That difference turns the economy into a capability.

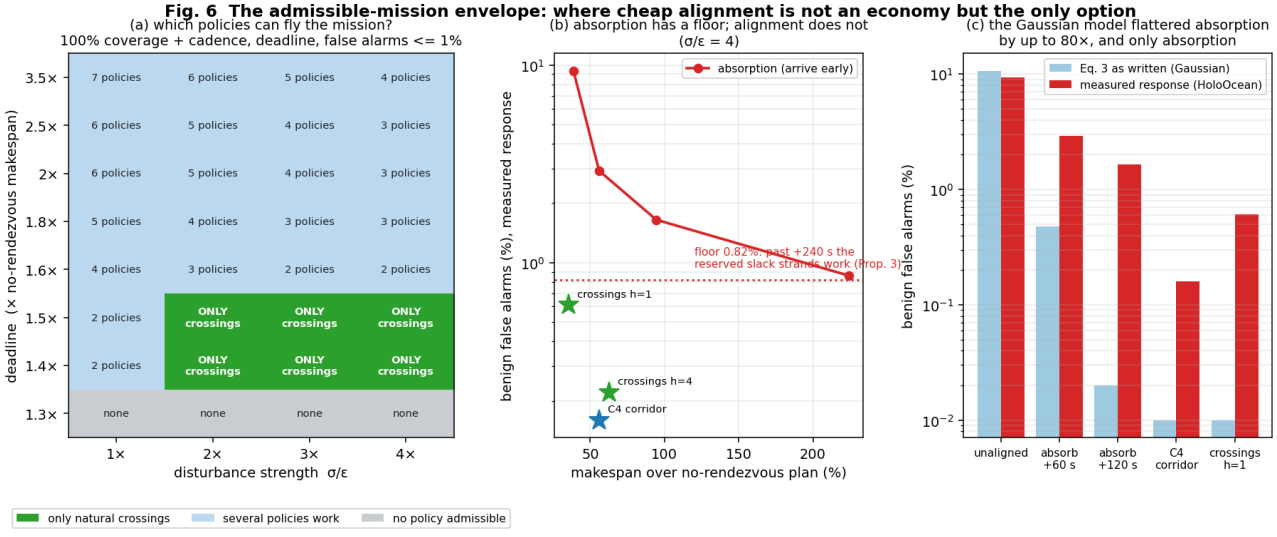


Fig. 6. (a) The admissible-mission envelope over deadline and disturbance, at a 1% false-alarm target. In the green cells the natural-crossing construction is the only policy that can fly the mission; above them several policies work, and below them the mesh itself does not fit the deadline. (b) Absorption's floor: past +240 s of reserved slack the schedule strands coverage work, so no budget reaches below 0.80%, while the aligned schedules sit an order of magnitude lower and cheaper. (c) The Gaussian form of Eq. (3) flattered absorption by up to 80× and left alignment essentially unchanged, which is why the model validation had to precede the comparison.

Fig. 6(a) maps it. At a generous deadline every policy is admissible and the choice is an economy. Tightening the deadline to 1.5× the no-rendezvous makespan removes absorption and the corridor together — both cost more than +50% — and at a disturbance of 2× or more

it also removes the unaligned mesh, which breaches the false-alarm target. What is left is the crossing construction alone. Tightening further, to 1.3×, removes everything: the mesh has an irreducible cost of about +35% and no rendezvous schedule fits inside that, which is an honest boundary on the claim rather than a result. The exclusive region is therefore narrow but real, and it is exactly the operating point a deadline-driven survey sits at.

TABLE IV — The exclusive region persists with team size: makespan overhead (%) / measured false alarms (%) at σ/ε = 4, three instances each.

K, N	unaligned	absorb +60 s	absorb +240 s	C4 corridor	crossings h=1
4, 90	38.7 / 9.92	56.0 / 3.01	224 / 0.84	56.0 / 0.00	35.4 / 0.73 yes
6, 120	25.9 / 5.42	52.9 / 1.89	187 / 0.45	52.9 / 0.00	35.5 / 0.42 yes
8, 160	28.6 / 4.81	52.2 / 1.56	192 / 0.28	52.2 / 0.00	33.8 / 0.63 yes

"yes" marks the policies admissible at a deadline of 1.5× with a 1% false-alarm target. In every team size the crossing construction is the only one, and it is the only column whose cost does not grow with the protection it provides.

Two features of Table IV are worth separating. The crossing construction is admissible at all three team sizes, and it is the only one; but note also that it is the only column whose overhead is flat in K (35.4, 35.5, 33.8) while remaining protected. Absorption's overhead is flat too, but its false alarms are an order of magnitude worse; the corridor matches the protection but never fits the deadline. We do not claim the exclusive region widens with K — on these three sizes it does not measurably — only that it does not close.

F. The look-ahead frontier

TABLE V — How far ahead anchors are reserved traces the whole cost-versus-guarantee frontier (four instances, σ/ε = 4).

look-ahead h	0	1	2	3	4	6
makespan +%	38.4	35.5	42.1	62.5	63.2	123.7
FA realised %	0.10	0.02	0.00	0.00	0.00	0.00
no-slack bound %	26.3	12.8	10.1	3.9	0.2	0.9
approach (s)	59	37	32	22	16	20
fell back to dwell %	45	20	16	6	0	2

Reserving one round ahead is the knee: it removes most of the fallbacks at no cost. Reserving four rounds ahead buys the full separation-only-grade bound (0.2%) at +63.2%, which is what the corridor costs — so the guarantee is available either way, and the crossing route to it is not cheaper. Reserving six rounds ahead is worse on **both** axes, because the locked anchors starve the routes. The knob is genuinely two-sided and the optimum is interior.

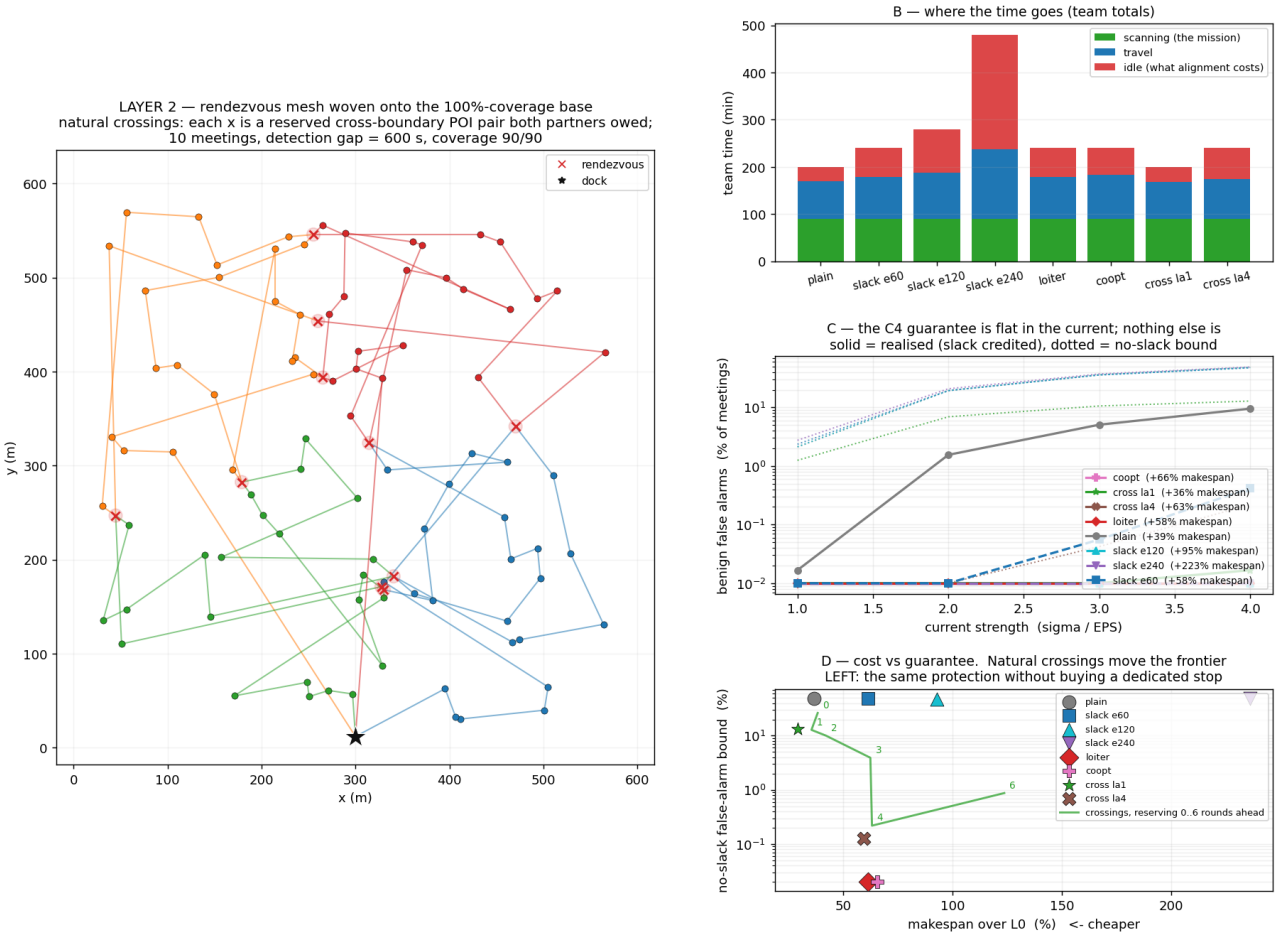


Fig. 4. (a) A woven mesh on the 100%-coverage base; each × is a reserved cross-boundary task pair that both partners owed. (b) Where the time goes: the corridor's cost is the red idle band. (c) False alarms against current strength, solid realised and dotted no-slack. (d) The cost-versus-guarantee plane, with the crossing frontier annotated by look-ahead depth. Legend keys map to Table II as: plain = unaligned mesh, slack eNN = arrive early +NN s, loiter = C4 corridor, coopt = C4 corridor task-sited, cross laH = crossings h = H.

G. Negative and null results

We record four findings that did not go our way, because each of them would otherwise be re-attempted.

- **Task-siting the corridor does not pay.** Placing the corridor meeting on a task while retaining the dwell budget costs more than it saves (+63.6% against +58.3%): the route distortion exceeds the recovered idle. An early single-instance result suggested otherwise and did not survive five instances.
- **Tightening α is free, which contradicts the expected Pareto.** Sweeping α from 0.4 to 0.02 shrinks d* from 6.8 m to 0.9 m with the corridor's makespan flat at +58%. The binding cost is the stop, not the corridor radius, so the false-alarm-versus-travel trade-off that the stylized proxy predicts does not exist on real routes — the cost is a step, not a curve.
- **Crossings at h = 1 do not deliver the separation-only guarantee.** Their no-slack bound is 11.5% and they satisfy $d \le d^*$ on no appointment; their low measured rate leans partly on the schedule's own slack. Reserving four rounds ahead buys the strict bound, at the corridor's price. The honest product is the frontier of Table V, not a single winner.
- **Multi-witness confirmation is useless here**, as the correlated model predicts: agreeing silent neighbours are exculpatory rather than corroborating. We report it because it is the first thing an i.i.d. intuition suggests.

One further negative is methodological and we think it is the most useful thing in this section. An earlier version of this work concluded, from Eq. (3) as written, that simply arriving early was competitive with alignment and that the separation-only formulation was therefore not obviously worth its cost. That conclusion was an artefact of a Gaussian assumption in a model we had authored ourselves, and it survived until the model was checked against an independent physics engine. It is worth stating plainly because the failure mode generalizes: a self-authored noise model will tend to be symmetric, symmetric noise flatters any policy that buys a margin, and buying a margin is what most engineering fixes do.

VII. DISCUSSION AND LIMITATIONS

The practical reading of Theorem 1 is that alignment should be bought in the currency that is cheap. Separation is expensive on a coverage route because it is dragged out of the geometry; duration is cheap because it is a by-product of the order in which tasks are visited, and ordering is free. The crossing construction is simply the cheapest point of A that a coverage route can reach without leaving its own work.

Several limitations bound the claims. The results are simulation only — no field trials — though Section VI-A tests the accrual model against an independent physics engine rather than against our own kinematics. That test ran on HoloOcean's HoveringAUV because its SurfaceVessel does not respond to ocean currents at all, so free-surface effects on a real USV remain unexamined, and it covers one vehicle and one controller. The disturbance is a stationary Gaussian field with a known correlation length, and robustness to a mis-estimated λ is untested — the safe direction is to use a lower bound on λ, which shrinks both corners. We model a single failure; the recovery table is indexed by one victim, and concurrent failures are outside it. The accrual model of (3) treats a completed scan as a perfect resynchronization point, which is the assumption that makes the final approach the whole exposure; a model in which delay accumulates across an entire window would move both corners and we would expect it to favour the crossing corner less. The measured response is convex, so the absolute false-alarm figures are optimistic (Section VI-A). We have not forward-executed a recovery **with** the mesh in place, so the interaction between a takeover and a reserved anchor whose owner has died is unmodelled. Finally, the alignment condition governs false alarms only; detection latency is governed by C3 and is independent of everything in Section IV.

VIII. CONCLUSION

We studied comms-free failure detection for a deadline-constrained multi-robot coverage team whose only evidence is a rendezvous that did not happen, under a disturbance whose spatial correlation makes a missed rendezvous ambiguous. We showed that the alignment requirement is a joint condition on approach duration and approach separation, that the separation-only constraint used previously is a conservative corner of it, and that letting duration co-vary with separation relaxes the admissible separation from $\Theta(B)$ to $\Theta(B^{(1/3)})$ — a factor that grows as the budget tightens and is 39× at our operating point. The construction that realizes this, siting each rendezvous on a reserved pair of adjacent tasks, costs no more than an unaligned mesh that offers no protection at all.

Testing the accrual model against an independent physics engine changed what that buys. Proportionality to exposure time and inheritance of the field's correlation both hold; linearity does not, and recomputing through the measured response shows that absorbing delay by arriving early — the obvious alternative, and one the Gaussian model makes competitive — has a false-alarm floor of 0.80% set by the cadence bound, while alignment has none. The result is a capability rather than an economy: there is a region of the (deadline, disturbance) plane, at a deadline of 1.5× the no-rendezvous makespan and a disturbance of 2× and above, in which our construction is the only policy that covers every task, meets the cadence, finishes on time and stays under a 1% false-alarm target, at team sizes 4, 6 and 8. Future work is to retire the single-failure assumption, to test sensitivity to a mis-estimated

correlation length, and to take the mesh to field trials.

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Reproducibility. Every number in Tables I–IV is produced by visualization/taskalloc_coverage_first.py in the accompanying repository (--dry_run runs the self-tests, --l2 --sweep regenerates the tables and Fig. 4); Fig. 2 and Fig. 3 by paper/make_paper_figs.py. Section VI-A and Fig. 5 come from visualization/holoocean_eq3_validation.py, run inside a HoloOcean environment (--exp probe, --exp time, --exp field --amp A), with the figure assembled by paper/make_eq3_fig.py. The disturbance physics is imported from the earlier prototype rather than reimplemented, and the self-tests assert that the closed form of Proposition 1 reproduces it exactly.